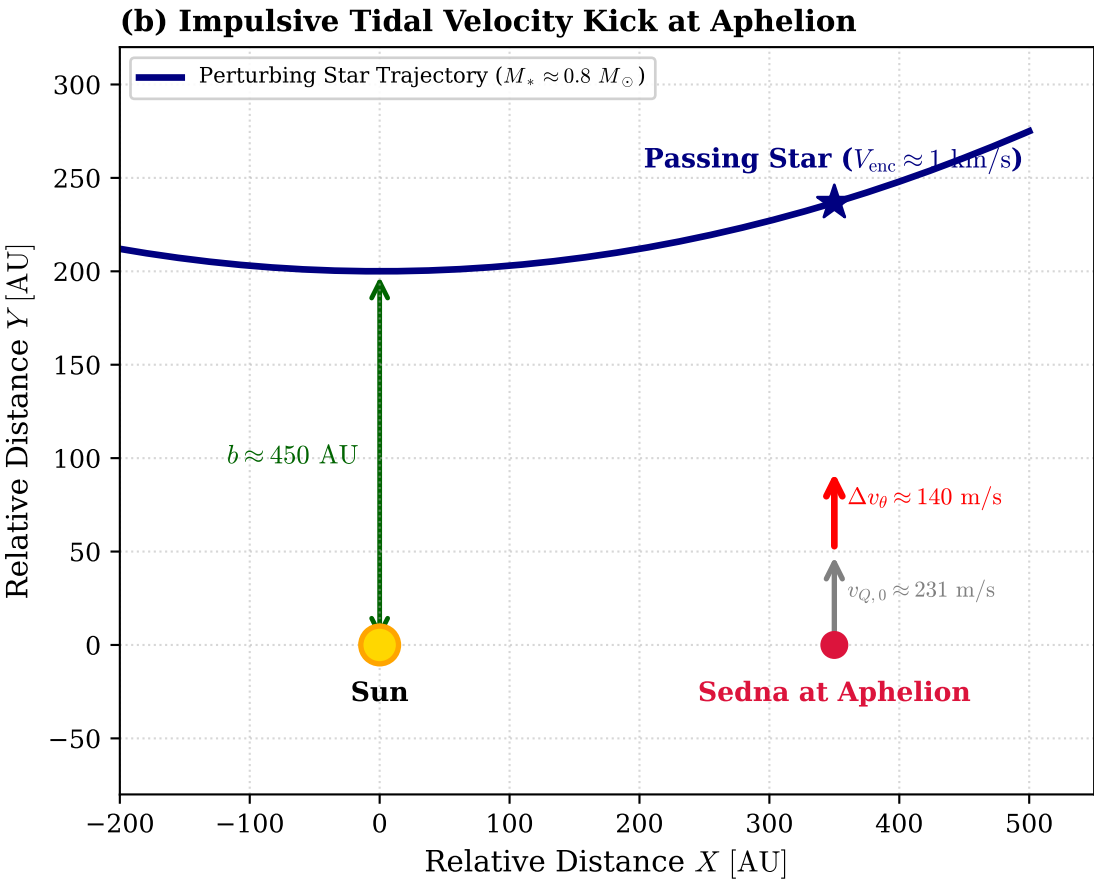
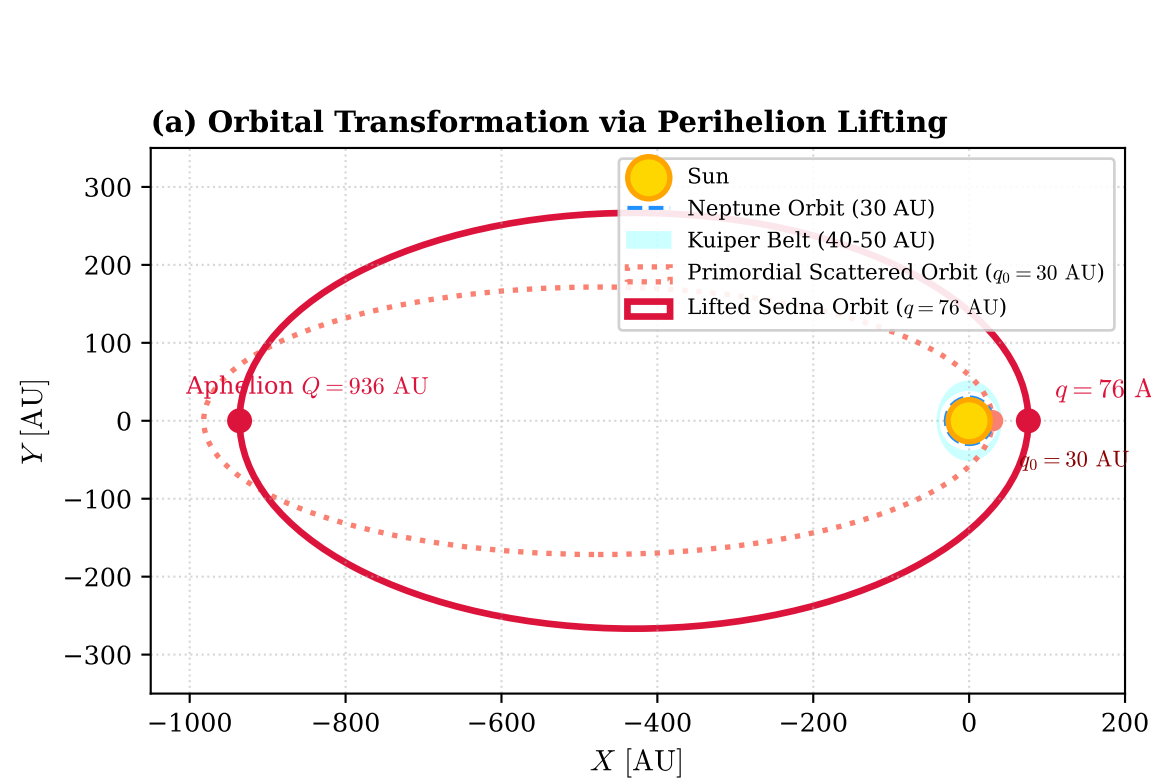


Conceptual Architecture & Stellar Encounter Dynamics



(c) Comprehensive Evaluation of Proposed Sedna Origin Hypotheses

Origin Mechanism	Key Physics / Theory	Predicted Perihelion q	Alignment varpi_i	Assessment / Status
1. Open Birth Cluster Flyby (Favored)	Impulsive stellar flyby ($b \sim 400\text{-}600$ AU)	$40 \text{ AU} < q < 100 \text{ AU}$ (Detached)	Random / isotropic	Highly probable (P $\sim 90\%$ in birth cluster)
2. Hypothetical Planet Nine	Secular Kozai-Lidov resonance torque	q oscillates between 35-85 AU	Clustered ($\Delta \text{varpi}_i \sim 1^\circ$)	Explains clustering of multi-eTNOs
3. Extra-Solar Planetesimal Capture	Tidal transfer from passing circumstellar disk	Broad q , high $a > 500$ AU	Isotropic / high i	Requires low capture cross-section
4. Massive Primordial Disk Self-Gravity	Secular collective disc eigenmode	q lifted by self-gravitating disc	Precessing eigenmodes	Requires unphysically massive disc